MAHATMA GANDHI UNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG4DSCECC200 – Electronics Service Technology

Duration:1.5 Hrs. **Maximum Marks:50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Multiple Choice Questions

Answer all questions.Each question carries 1 mark

1.	Which color represents the earth wire in domestic wiring?		U	CO1
	A. Red	B. Blue		
	C. Black	D. Green		
2.	A fuse primarily protects a circuit from		U	CO1
	A. Voltage fluctuation	B. Overload and short circuit		
	C. Power loss	D. Earthing failure		
3.	The function of an MCB is to		U	CO1
	A. Measure power	B. Control voltage		
	C. Protect against overcurrent	D. Improve power factor		
4.	Which instrument is used to measure AC current without breaking the circuit?		U	CO2
	A. Ammeter	B. Voltmeter		
	C. Clamp meter	D. LCR meter		
5.	RCCB mainly protects against		U	CO1
	A. Overload	B. Short circuit		
	C. Earth leakage current	D. Overvoltage		
6.	The output of a thermocouple is based on		U	CO2
	A. Resistance change	B. Capacitance change		
	C. Seebeck effect	D. Hall effect		
7.	A strain gauge works on the principle of		U	CO2
	A. Capacitance change	B. Resistance change		
	C. Inductance change	D. Voltage division		

8.	Identifying a faulty FET using a multimeter mainly involves checking A. Color code B. Gate insulation resistance C. Drain current rating D. Package type	A	CO3
9.	A wiring diagram mainly helps in A. Circuit analysis B. Component testing C. Installation and maintenance D. PCB design	A	CO3
10.	Precision testing of capacitors is done using A. Voltmeter B. Ammeter C. LCR meter D. Clamp meter	U	CO2

1 x 10 = 10

Part B

Short Answer Questions

Answer **4** Questions. Each question carries **5** marks

11.	What is the function of earthing in electrical systems?	U	CO1
12.	Explain the working principle of an MCB.	U	CO2
13.	Explain the working of an RTD.	A	CO3
14.	How can a diode be tested using a multimeter?	A	CO3
15.	Justify the need for both block and circuit diagrams in troubleshooting.	A	CO3
16.	Explain the importance of load calculation.	U	CO1

4 x 5 = 20

Part C

Essay Questions

Answer any **2** Questions. Each question carries **10 m**arks

17.	Explain the significance of wiring color codes.	U	CO1
18.	Explain the function of Current Transformer (CT) in power systems.	U	CO2
19.	Analyze the effects of poor earthing and harmonics on power quality and equipment performance.	U	CO1
20.	Describe continuity testing and its importance in PCB inspection.	A	CO3

2 x 10 = 20